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Progress towards and barriers to implementation of a risk framework for US federal wildland fire policy and decision making

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Wildland fire risk research in Canada

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Lynn M. Johnston, Xianli Wang, ... Mike D. Flannigan

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Ryan A. Fitch, Yeon Su Kim, ... Joe E. Crouse

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Wildland fire risk and social vulnerability in the Southeastern United States: An exploratory spatial data analysis approach

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Shawn P. Urbanski, Wei Min Hao, Stephen Baker

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Global wildland fire season severity in the 21st century

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Mike Flannigan, Alan S. Cantin, ... Lynn M. Gowman

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William J. de Groot, B. Michael Wotton, Michael D. Flannigan

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Information systems in support of wildland fire management decision making in Canada

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B. S Lee, M. E Alexander, ... P Englefield

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The impact of wildland fires on calcareous Mediterranean pedosystems (Sardinia, Italy) – An integrated multiple approach

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Gian Franco Capra, Simona Tidu, ... Peter Filzmoser

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Physics-informed neural networks for parameter learning of wildfire spreading

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Modeling topographic influences on fuel moisture and fire danger in complex terrain to improve wildland fire management decision support

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Modeling wildland fire propagation with level set methods

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Foliar moisture content of Pacific Northwest vegetation and its relation to wildland fire behavior

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Gavin R. McMeeking, Sonia M. Kreidenweis, ... Jeffrey L. Collett

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Klaus Braun

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Impact of underground wildfire on plant roots in peat soil

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